NutriClass: Food Classification Using Nutritional Data

Introduction

1. Developed a Machine Learning-based system to classify food items using nutritional attributes for promoting digital health awareness.

2. Applied all Supervised Model (Logistic, Decision Tree,KNN,Gradient ,XG Boost,Random Forest and SVM) for solving classification problem.

3. Performing data preprocessing and EDA(Exploratory Data Analysis) for better analysis.

4. Evaluated models using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix, achieving over 90% accuracy.

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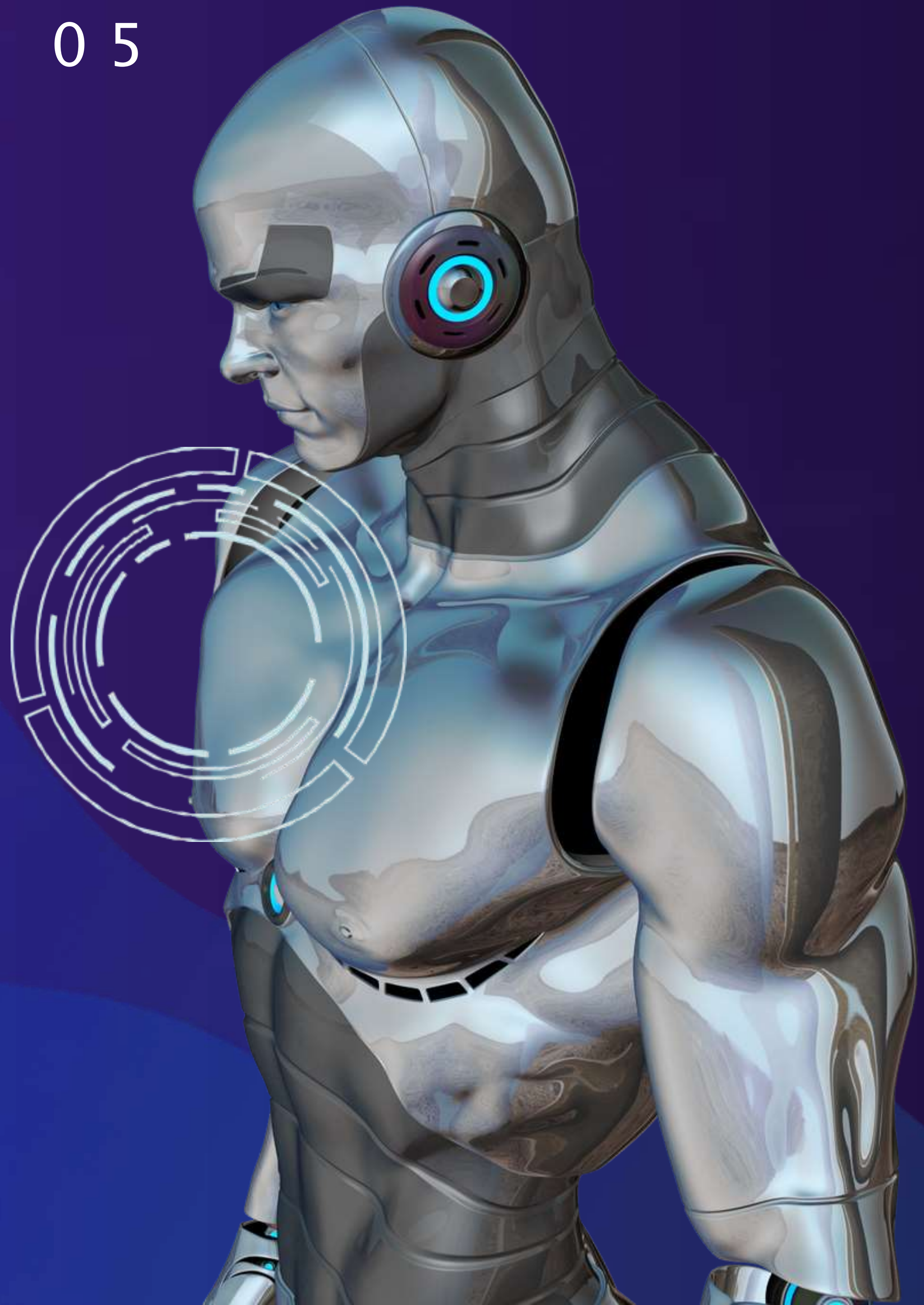

Objective

1. To analyze nutritional data using Exploratory Data Analysis (EDA).

2. To build and train the accuracy of every Supervised ML model for better analysis.

3. Reduce the outlier of the dataset for accurate ranging.

4. To evaluate and compare model performance to solve the real world problem.

Tool Used

1. Numpy : NumPy enabled efficient handling of multi-dimensional arrays, mathematical computations, and data transformations required during scaling, normalization, and feature engineering.

2. Pandas : Pandas was extensively used for data loading, cleaning, and manipulation.

3. Matplotlib : Matplotlib was used to create box plots, histograms, and performance graphs, helping visualize nutritional distributions and identify outliers.

4. Seaborn : It was used to create advanced visualizations such as KDE plots and confusion matrix heat maps.

5. Scikit-learn : Scikit-learn was used for feature scaling, imputation, encoding, train-test splitting, and implementing machine learning models .

Model Used

Logistic Regression

1. Logistic Regression is a linear classification model used as a baseline in this project.
2. It predicts food categories based on probability using nutritional features.
3. It helps understand whether the data is linearly separable.

Decision Tree

1. Decision Tree classifies food items using rule-based splitting of features.
2. It creates a tree structure basedon nutritional conditions like fat or calories.
3. It is easy to interpret and useful for understanding decision logic.

Random Forest

1. Random Forest is an ensemble of multiple decision trees.
2. It improves accuracy by combining predictions using majority voting.
3. It reduces overfitting and provides better stability than a single tree.

K-Nearest Neighbors (KNN)

1. KNN classifies food based on similarity to nearby data points.
2. It uses distance metrics to find the closest nutritional profiles.
3. Feature scaling is important for accurate performance.

Support Vector Machine (SVM)

1. SVM finds the optimal boundary that separates food categories.
2. It works well with high-dimensional nutritional data.
3. It maximizes the margin between different classes.

Gradient Boosting

1. Gradient Boosting builds models sequentially to correct previous errors.
2. It focuses more on misclassified food samples.
3. It improves prediction accuracy through boosting technique.

XGBoost

1. XGBoost is an advanced and optimized boosting algorithm.
2. It handles complex feature interactions and prevents overfitting using regularization.
3. It achieved the highest performance in this project.

FUTURE SCOPE

1. The dataset can be expanded to include more food categories and real-world nutritional data for better generalization.

2. Advanced models such as Deep Learning and Neural Networks can be implemented to improve classification accuracy.

3. The system can be integrated into a mobile or web application for real-time food classification and dietary recommendations.

4. Personalized nutrition features can be added based on user health conditions, age, and dietary preferences.

Thank You!

We welcome your feedback and questions

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